Writing Equations

Chemists write equations to represent what is happening in a chemical reaction. Equations are useful because they provide an easy way to show complex reactions.

Word Equations

- The simplest form of chemical equation is a word equation.
- The most general word equation is:

  Reactants → products
- This general equation can then be expanded to:

  Substance A + Substance B → Substance C + Substance D
- For example, consider the reaction between magnesium and hydrochloric acid. This reaction produces magnesium chloride and hydrogen gas. It can be written as a word equation.

  Reactants → products

  Magnesium + Hydrochloric acid → Magnesium chloride + Hydrogen gas

Example 1: Lead reacts with oxygen gas to produce lead (IV) oxide. Write a word equation to describe the reaction.

Using Formulae

- A chemical equation tells you even more information if it includes the element symbols and chemical formulae of all the substances involved.
- In the example above, the reactants and products can be written using symbols for the elements and compounds.
- The reaction can be written as:

  $Mg + HCl \rightarrow MgCl_2 + H_2$

Rules for Formula Equations

- Metals are written as single atoms: copper is written as Cu and iron is written as Fe
- Non-metal gases are written as diatomic molecules: oxygen is written as O_2, hydrogen is written as H_2, chlorine is written as Cl_2 etc.

Example 2: For the reaction in example 1, write a formula equation.

Example 3: Write the formula equation for the following word equation:

Carbon + Oxygen gas → Carbon dioxide

Questions

1. Write **word equations** for the following reactions.
    a. When sulphuric acid is added to iron metal, hydrogen gas and iron (II) sulphate are formed.

    b. If sulphuric acid is poured onto solid sodium carbonate, bubbles of carbon dioxide are produced, as well as water and sodium sulphate.

    c. Magnesium burns easily in oxygen, producing magnesium oxide.

    d. An iron nail exposed to the oxygen in air will rust, forming iron oxide.

    e. When solutions of lead nitrate and sodium iodide are mixed, a yellow precipitate of lead iodide is formed, as well as sodium nitrate in solution.

2. For each of the word equations, rewrite it using the **chemical formulae**.